MA1508E: Linear Algebra for Engineering

7. System of Linear Differential Equations

7.1 First Order Linear System of Differential Equations

First Order Linear System of Differential Equations

A first order linear system of differential equations (with variable t) in its *standard form* is given by

$$\begin{cases} y_1'(t) & = & a_{11}(t)y_1(t) & + & \cdots & + & a_{1n}(t)y_n(t) & + & g_1(t) \\ y_2'(t) & = & a_{21}(t)y_1(t) & + & \cdots & + & a_{2n}(t)y_n(t) & + & g_2(t) \\ & \vdots & & & & & & & \\ y_n'(t) & = & a_{n1}(t)y_1(t) & + & \cdots & + & a_{nn}(t)y_n(t) & + & g_n(t) \end{cases}.$$

This system can be rewritten in matrix form as

$$\begin{pmatrix} y_1'(t) \\ y_2'(t) \\ \vdots \\ y_n'(t) \end{pmatrix} = \begin{pmatrix} a_{11}(t) & \cdots & a_{1n}(t) \\ a_{21}(t) & \cdots & a_{2n}(t) \\ \vdots & \ddots & \vdots \\ a_{n1}(t) & \cdots & a_{nn}(t) \end{pmatrix} \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix} + \begin{pmatrix} g_1(t) \\ g_2(t) \\ \vdots \\ g_n(t) \end{pmatrix}.$$

In compact vector notation, this is written as

$$\mathbf{y}'(t) = \mathbf{A}(t)\mathbf{y}(t) + \mathbf{g}(t).$$

Here, $\mathbf{A}(t)$ is the coefficient matrix, which is generally a matrix-valued function of t. $\mathbf{y}(t)$ is the vector of unknown functions, $\mathbf{A}(t)$ governs the system dynamics, and $\mathbf{g}(t)$ is a given forcing term.

First Order Linear System of Differential Equations

A first-order linear system of differential equations is given by

$$\mathbf{y}'(t) = \mathbf{A}(t)\mathbf{y}(t) + \mathbf{g}(t)$$

Depending on the properties of the system, we classify it as follows:

- **Homogeneous**: The system is called homogeneous if there is no forcing term, $\mathbf{g}(t) = 0$, reducing the equation to
$$\mathbf{y}'(t) = \mathbf{A}(t)\mathbf{y}(t).$$
- **Constant coefficient**: The system has constant coefficients if the entries of $\mathbf{A}(t)$, denoted by $\mathbf{a}_{ij}(t)$, are constants. In this case, $\mathbf{A}$ is a real-valued constant matrix.

In this course, we will focus exclusively on homogeneous first-order linear systems with constant coefficients, which take the standard form:

$$\mathbf{y}'(t) = \mathbf{A}\mathbf{y}(t), \quad \text{where} \quad \mathbf{A} = (a_{ij})_{n\times n}, \quad a_{ij} \in \mathbb{R}.$$

Examples

1. Non-homogeneous, non-constant coefficient first-order system:

$$\begin{cases} y_1'(t) & = & \cos(t)y_1(t) & - & \sin(t)y_2(t) & + & t^2 \\ y_2'(t) & = & \sin(t)y_1(t) & + & \cos(t)y_2(t) & + & 2t \end{cases}$$

   In matrix form:

$$\begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}' = \begin{pmatrix} \cos(t) & -\sin(t) \\ sin(t) & \cos(t) \end{pmatrix} \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix} + \begin{pmatrix} t^2 \\ 2t \end{pmatrix}.$$

2. Homogeneous, non-constant coefficient first-order system:

$$\begin{cases} y_1'(t) & = & \cos(t)y_1(t) & - & \sin(t)y_2(t) \\ y_2'(t) & = & \sin(t)y_1(t) & + & \cos(t)y_2(t) \end{cases}$$

   In matrix form:

$$\begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}' = \begin{pmatrix} \cos(t) & -\sin(t) \\ \sin(t) & \cos(t) \end{pmatrix} \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}.$$

Examples

3. Homogeneous, constant coefficient first-order system:

$$\begin{cases} y_1'(t) & = & y_1(t) & + & y_2(t) \\ y_2'(t) & = & 2y_1(t) & - & y_2(t) \end{cases}$$

In matrix form:

$$\begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}' = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}.$$

Solutions to System of Different Equations

A function-valued vector $\mathbf{x}(t)$ is called a solution to the (first-order homogeneous linear) differential system with constant coefficients:

$$\mathbf{y}'(t) = \mathbf{A}\mathbf{y}(t)$$

if it satisfies

$$\mathbf{x}'(t) = \mathbf{A}\mathbf{x}(t).$$

An initial condition specifies the value of the solution at some initial time t_0:

$$\mathbf{y}(t_0) = \begin{pmatrix} y_1(t_0) \\ \vdots \\ y_n(t_0) \end{pmatrix} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{R}^n.$$

A function $\mathbf{x}(t)$ is a solution to the system with this initial condition if

$$\mathbf{x}(t_0) = \mathbf{a}.$$

A system of differential equations together with an initial condition is known as an initial value problem (IVP). Solving an IVP means finding a function $\mathbf{x}(t)$ that satisfies both the differential equation and the given initial condition.

Example

Consider the differential system

$$\begin{cases} y_1'(t) &= y_1(t) \\ y_2'(t) &= \qquad\qquad 2y_2(t) \end{cases}$$

with initial condition $y_1(0) = 1 = y_2(0)$.

The system can be rewritten as: $\begin{pmatrix} y_1'(t) \\ y_2'(t) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}$.

Let $\mathbf{x}(t) = \begin{pmatrix} e^t \\ e^{2t} \end{pmatrix}$. Since

$$\mathbf{x}'(t) = \begin{pmatrix} e^t \\ 2e^{2t} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} e^t \\ e^{2t} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \mathbf{x}(t),$$

$\mathbf{x}(t)$ satisfies the differential equation. Moreover, since $\mathbf{x}(0) = \begin{pmatrix} e^0 \\ e^{2(0)} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ satisfies the given initial condition, we conclude that $\mathbf{x}(t) = \begin{pmatrix} e^t \\ e^{2t} \end{pmatrix}$ is a solution to the initial value problem.

Remarks

- The solution from the previous example can be rewritten as

$$\mathbf{x}(t) = e^t \begin{pmatrix} 1 \\ 0 \end{pmatrix} + e^{2t} \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

- Notice that $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are eigenvectors of the coefficient matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

  corresponding to the eigenvalues 1 and 2, respectively.

- This property holds for all first-order homogeneous linear systems of differential equations (SDEs) with constant coefficients: Linear combinations of exponential functions multiplied by the corresponding eigenvectors of $\mathbf{A}$ are solutions to the differential system.

Solutions to System of Different Equations

Theorem

Suppose $\mathbf{v}$ *is an eigenvector associated with the eigenvalue* λ *of a matrix* $\mathbf{A}$*. Then the function*

$$\mathbf{x}(t) = \mathbf{v}e^{\lambda t}$$

is a solution to the first-order homogeneous linear system of differential equations with constant coefficients:

$$\mathbf{y}' = \mathbf{A}\mathbf{y}.$$

Proof.

Since λ is an eigenvalue of $\mathbf{A}$ with corresponding eigenvector $\mathbf{v}$, we have $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$. Define $\mathbf{x}(t) = \mathbf{v}e^{\lambda t}$. Then differentiating both sides gives

$$\mathbf{x}'(t) = \frac{d}{dt}\left(\mathbf{v}e^{\lambda t}\right) = \lambda\mathbf{v}e^{\lambda t} = \mathbf{A}\mathbf{v}e^{\lambda t} = \mathbf{A}\mathbf{x}(t).$$

Thus, $\mathbf{x}(t) = \mathbf{v}e^{\lambda t}$ satisfies the differential equation, proving that it is a solution. $\square$

Example

Consider the differential system

$$\begin{cases} y_1' &= y_1 \\ y_2' &= y_1 + 2y_2 \end{cases}$$

The coefficient matrix is $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}$.

Verify that $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is an eigenvector of $\mathbf{A}$ associated to eigenvalue $\lambda = 1$.

Let $\mathbf{x}_1(t) = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t$. Then

$$\begin{aligned} \mathbf{x}_1'(t) &= \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t \\ \mathbf{A}\mathbf{x}_1(t) &= \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t \end{aligned}$$

This shows that $\mathbf{x}_1(t) = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t$ is a solution to the differential system.

Example

Consider the differential system

$$\begin{cases} y_1' &= y_1 \\ y_2' &= y_1 + 2y_2 \end{cases}$$

Verify that $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is another eigenvector of $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}$ associated to eigenvalue $\lambda = 2$.

So, $\mathbf{x}_2(t) = \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{2t}$ is another solution to the SDE. Further, for any real numbers α and β,

$$\begin{aligned} \frac{d}{dt}\left(\alpha \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{t} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{2t}\right) &= \alpha \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{t} + 2\beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{2t} \\ \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}\left(\alpha \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{t} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{2t}\right) &= \alpha \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{t} + \beta \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{2t} = \alpha \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{t} + 2\beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{2t} \end{aligned}$$

which shows that $\left(\alpha \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{t} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{2t}\right)$ is also a solution to the SDE.

This demonstrates that linear combinations of solutions to a first order homogeneous linear SDE with constant coefficient is a solution too.

Solutions to System of Different Equations

Theorem (Superposition principle)

Suppose $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are solutions to $\mathbf{y}' = \mathbf{A}\mathbf{y}$. For any $\alpha, \beta \in \mathbb{R}$,

$$\alpha\mathbf{x}_1(t) + \beta\mathbf{x}_2(t)$$

is also a solution to the system of differential equations.

Proof.

This follows from the linearity of both differentiation and matrix multiplication.

Since $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are solutions, we have

$$\mathbf{x}_1'(t) = \mathbf{A}\mathbf{x}_1(t) \quad \text{and} \quad \mathbf{x}_2'(t) = \mathbf{A}\mathbf{x}_2(t).$$

Let $\mathbf{x}(t) = \alpha\mathbf{x}_1(t) + \beta\mathbf{x}_2(t)$. Then

$$\mathbf{A}\mathbf{x}(t) = \mathbf{A}(\alpha\mathbf{x}_1(t) + \beta\mathbf{x}_2(t)) = \alpha\mathbf{A}\mathbf{x}_1(t) + \beta\mathbf{A}\mathbf{x}_2(t) = \alpha\mathbf{x}_1'(t) + \beta\mathbf{x}_2'(t) = \mathbf{x}'(t).$$

$\square$

Exercise

Let $\mathbf{A} = \begin{pmatrix} 4 & -5 \\ -2 & 1 \end{pmatrix}$. Consider the initial value problem $\mathbf{y}'(t) = \mathbf{A}\mathbf{y}(t)$ with initial condition $\mathbf{x}(0) = \begin{pmatrix} 2.9 \\ 2.6 \end{pmatrix}$.

- Check that $\lambda_1 = 6$ and $\lambda_2 = -1$ are eigenvalues of $\mathbf{A}$, with corresponding eigenvectors $\begin{pmatrix} -5 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.
- Use the corresponding eigenvalues and eigenvectors above to form 2 solutions $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$.
- Let $\mathbf{x}(t) = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t)$ for some real numbers c_1, c_2.
- If $\mathbf{x}(t)$ is a solution to the initial value problem, then it must be that $\mathbf{x}(0) = \begin{pmatrix} 2.9 \\ 2.6 \end{pmatrix}$. Use this to form a linear system with variables c_1, c_2, and solve for c_1 and c_2.
- You should arrive at
$$\mathbf{x}(t) = \frac{-3}{70}\begin{pmatrix} -5 \\ 2 \end{pmatrix} e^{6t} + \frac{188}{70}\begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t}.$$
- Check that this is indeed the solution to the initial value problem.

Remarks

- Consider the differential system $\mathbf{y}'(t) = \mathbf{A}\mathbf{y}(t)$.
- Note that the trivial solution, $\mathbf{x}(t) = \mathbf{0}$ for all t, satisfies this system.
- By the superposition principle, the set of all solutions forms a subspace (within some ambient vector space, which we do not explicitly specify).
- Consequently, the unique solution to an initial value problem must be an element of this subspace.
- This naturally leads to the question: is there a basis for the solution space?
- This was also hinted to in the previous exercise, we implicitly assumed such a basis by expressing the unique solution as a linear combination of two particular solutions.
- To formalize this idea, we now introduce a few key definitions.

Linear Independence of Solutions

A function-valued vector $\mathbf{v}(t)$ is zero if for every t, $\mathbf{v}(t) = \mathbf{0} \in \mathbb{R}^n$.

A set $\{\mathbf{x}_1(t), \mathbf{x}_2(t), ..., \mathbf{x}_k(t)\}$ of function-valued vector is linearly independent if whenever $c_1, c_2, ..., c_k \in \mathbb{R}$ are real numbers such that

$$c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t) + \cdots + c_k\mathbf{x}_k(t) = \mathbf{0}$$

for all t in the domain, necessarily $c_1 = c_2 = \cdots = c_k = 0$. Otherwise, we say that it is linearly dependent.

Caution: For a function-valued vector $\mathbf{v}(t)$ to be zero, $\mathbf{v}(t) = 0$ for all $t \in \mathbb{R}$. Hence, for $\{\mathbf{x}_1(t), \mathbf{x}_2(t), ..., \mathbf{x}_k(t)\}$ to be linearly dependent, we can find some $c_1, c_2, ..., c_k$ not all zero, and a $t \in \mathbb{R}$ such that

$$c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t) + \cdots + c_k\mathbf{x}_k(t) = \mathbf{0}.$$

Example

Determine if the set $\{\mathbf{v}_1(t), \mathbf{v}_2(t), \mathbf{v}_3(t)\}$ is linearly independent, where

$$\mathbf{v}_1(t) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2(t) = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_3(t) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Note that these function-valued vectors are constant vectors, meaning they do not actually depend on t. Then, the set is linearly independent as a set of function-valued vector is equivalent to it being linearly independent as a set of real-valued vectors, that is,

$$c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

has only the trivial solution $c_1 = c_2 = c_3 = 0$.

Notice that even though these are function-valued vectors, since they are constants, the linear independence works exactly the same way as with regular vectors in $\mathbb{R}^3$.

Discussion

Is the set $\left\{ \begin{pmatrix} t \\ \cos(t) \\ e^t \end{pmatrix}, \begin{pmatrix} \sin(t) \\ t^2 \\ t \end{pmatrix} \right\}$ linearly independent?

Solve for c_1, c_2 such that

$$c_1 \begin{pmatrix} t \\ \cos(t) \\ e^t \end{pmatrix} + c_2 \begin{pmatrix} \sin(t) \\ t^2 \\ t \end{pmatrix} = \begin{pmatrix} t & \sin(t) \\ \cos(t) & t^2 \\ e^t & t \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} t & \sin(t) & 0 \\ \cos(t) & t^2 & 0 \\ e^t & t & 0 \end{array} \right) \xrightarrow{???}$$

cannot perform elementary row operation!

But if the coefficient matrix is a square matrix, we can use one of the equivalent statements of invertibility.

Wronskian

But in the special case where we have 3 vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ in $\mathbb{R}^3$, linear independence of the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is equivalent to the determinant of the matrix $\begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{pmatrix}$ being nonzero,

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \text{ is linearly independent} \quad \Leftrightarrow \quad \det\left(\begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{pmatrix}\right) \neq 0.$$

This inspires the following definition.

Definition

Let $S = \{\mathbf{x}_1(t), \mathbf{x}_2(t), ..., \mathbf{x}_n(t)\}$ be a set containing n functioned-valued vectors with n coordinates. Define the Wronskian of S to be

$$W(\mathbf{x}_1(t), \mathbf{x}_2(t), ..., \mathbf{x}_n(t)) = \det\left(\begin{pmatrix} \mathbf{x}_1(t) & \mathbf{x}_2(t) & \cdots & \mathbf{x}_n(t) \end{pmatrix}\right).$$

Then $S = \{\mathbf{x}_1(t), \mathbf{x}_2(t), ..., \mathbf{x}_n(t)\}$ is linearly independent if the Wronskian is not the constant zero function,

$$W(\mathbf{x}_1(t), \mathbf{x}_2(t), ..., \mathbf{x}_n(t)) \neq 0.$$

Example

Let $S = \left\{ \begin{pmatrix} \cos(t) \\ t \end{pmatrix}, \begin{pmatrix} \sin(t) \\ t \end{pmatrix} \right\}$. Then the Wronskian is

$$W\left(\begin{pmatrix} \cos(t) \\ t \end{pmatrix}, \begin{pmatrix} \sin(t) \\ t \end{pmatrix} \right) = \begin{vmatrix} \cos(t) & \sin(t) \\ t & t \end{vmatrix} = t(\cos(t) - \sin(t))$$

This set is linearly independent for if we take $t = 2\pi$, then the Wronskian is $2\pi \neq 0$.

Example

Let $S = \left\{ \begin{pmatrix} e^t \\ 2e^t \\ -e^t \end{pmatrix}, \begin{pmatrix} -e^t \\ 0 \\ e^t \end{pmatrix}, \begin{pmatrix} e^{2t} \\ e^{2t} \\ 0 \end{pmatrix} \right\}$. Then the Wronskian is

$$\begin{aligned} W\left(\begin{pmatrix} e^t \\ 2e^t \\ -e^t \end{pmatrix}, \begin{pmatrix} -e^t \\ 0 \\ e^t \end{pmatrix}, \begin{pmatrix} e^{2t} \\ e^{2t} \\ 0 \end{pmatrix} \right) &= \begin{vmatrix} e^t & -e^t & e^{2t} \\ 2e^t & 0 & e^{2t} \\ -e^t & e^t & 0 \end{vmatrix} \\ &= (e^t)(e^t)(e^{2t}) \begin{vmatrix} 1 & -1 & 1 \\ 2 & 0 & 1 \\ -1 & 1 & 0 \end{vmatrix} \\ &= 2e^{4t} \end{aligned}$$

Hence, the set is linearly independent since the Wronskian is $2 \neq 0$ when $t = 0$.

Caution

A set $S = \{\mathbf{x}_1(t), \mathbf{x}_2(t), ..., \mathbf{x}_n(t)\}$ of function-valued vectors can be linearly independent even though the Wronskian is the constant zero function. That is, if $W \neq 0$, we can conclude that S is linearly independent, but if S is linearly independent, it may be that $W = 0$.

For example, consider

$$S = \left\{ \begin{pmatrix} t \\ t \end{pmatrix}, \begin{pmatrix} t^2 \\ t^2 \end{pmatrix} \right\}.$$

Then S is linearly independent since

$$c_1 \begin{pmatrix} t \\ t \end{pmatrix} + c_2 \begin{pmatrix} t^2 \\ t^2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ for all } t$$

$$\Leftrightarrow t(c_1 + c_2 t) = 0 \text{ for all } t \Leftrightarrow c_1 = 0 = c_2.$$

But the Wronskian is

$$W\left(\begin{pmatrix} t \\ t \end{pmatrix}, \begin{pmatrix} t^2 \\ t^2 \end{pmatrix} \right) = \begin{vmatrix} t & t^2 \\ t & t^2 \end{vmatrix} = t^3 - t^3 = 0$$

for all t.

Fundamental set of solutions

Definition

A set $S = \{\mathbf{x}_1(t), \mathbf{x}_2(t), ..., \mathbf{x}_n(t)\}$ of solutions to the differential system $\mathbf{y}' = \mathbf{A}\mathbf{y}$ is called a fundamental set of solutions if its Wronskian is nonzero.

Let $S = \{\mathbf{x}_1(t), \mathbf{x}_2(t), ..., \mathbf{x}_n(t)\}$ be a fundamental set of solutions to a differential system $\mathbf{y}' = \mathbf{A}\mathbf{y}$. Then, the general solution to the differential system is

$$\mathbf{x}(t) = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t) + \cdots + c_n\mathbf{x}_n(t), \quad c_1, c_2, ..., c_n \in \mathbb{R}.$$

That is, any solution to the differential system $\mathbf{y}' = \mathbf{A}\mathbf{y}$ must be a linear combination of the function-valued vectors in the fundamental set of solutions.

Suppose $\mathbf{y}(t_0) = \mathbf{a} \in \mathbb{R}^n$ is an initial condition for the differential system, then we are able to find (the) solution to the initial value problem by solving for $c_1, c_2, ..., c_n$ in the equation

$$c_1\mathbf{x}_1(t_0) + c_2\mathbf{x}_2(t_0) + \cdots + c_n\mathbf{x}_n(t_0) = \mathbf{a}.$$

Fundamental Set of Solutions for Diagonalizable Coefficient Matrices

Theorem

Suppose $\mathbf{A}$ *is diagonalizable. Let* $\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_n$ *be* n *linearly independent eigenvectors associated to (real) eigenvalues* $\lambda_1, \lambda_2, ..., \lambda_n$ *(not necessarily distinct). Then*

$$\left\{\mathbf{v}_1 e^{\lambda_1 t}, \mathbf{v}_2 e^{\lambda_2 t}, ..., \mathbf{v}_n e^{\lambda_n t}\right\}$$

is a fundamental set of solutions to the first order homogeneous system of differential equations $\mathbf{y}' = \mathbf{A}\mathbf{y}$ *with constant coefficients, and*

$$\mathbf{x}(t) = c_1 \mathbf{v}_1 e^{\lambda_1 t} + c_2 \mathbf{v}_2 e^{\lambda_2 t} + \cdots + c_n \mathbf{v}_n e^{\lambda_n t}, c_1, c_2, ..., c_n \in \mathbb{R}$$

is the general solution.

Proof.

Exercise. □

Example

Solve the differential system

$$\begin{cases} y_1' &= y_1 \\ y_2' &= y_1 + 2y_2 \end{cases}$$

with initial conditions: $y_1(0) = 1$, $y_2(0) = 1$.

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}, \text{ eigenvalues are } \lambda = 1, 2.$$

$\lambda = 1$: $\mathbf{I} - \mathbf{A} = \begin{pmatrix} 0 & 0 \\ -1 & -1 \end{pmatrix}$. So $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is an associated eigenvector.

$\lambda = 2$: $2\mathbf{I} - \mathbf{A} = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}$. So $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is an associated eigenvector.

$$\Rightarrow \left\{ e^t \begin{pmatrix} 1 \\ -1 \end{pmatrix}, e^{2t} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \text{ is a fundamental set of solutions.}$$

Example

Wronskian is

$$W\left(\begin{pmatrix} e^t \\ -e^t \end{pmatrix}, \begin{pmatrix} 0 \\ e^{2t} \end{pmatrix}\right) = \begin{vmatrix} e^t & 0 \\ -e^t & e^{2t} \end{vmatrix} = e^{3t} \neq 0.$$

$$\Rightarrow \text{General solution: } \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = c_1 e^t \begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

or, $y_1 = c_1 e^t$ and $y_2 = -c_1 e^t + c_2 e^{2t}$. Initial condition

$$\begin{array}{ccccccc} 1 & = & y_1(0) & = & c_1 & & \\ 1 & = & y_2(0) & = & -c_1 & + & c_2 \end{array}, \left(\begin{array}{cc|c} 1 & 0 & 1 \\ -1 & 1 & 1 \end{array}\right) \to \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right)$$

$\Rightarrow c_1 = 1$ and $c_2 = 2$. So solution to the initial value problem is

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} e^t \\ -e^t + 2e^{2t} \end{pmatrix},$$

or $y_1 = e^t$ and $y_2 = 2e^{2t} - e^t$.

Example

Solve the differential system

$$\begin{cases} y_1' &= 3y_1 &+ y_2 &- y_3 \\ y_2' &= y_1 &+ 3y_2 &- y_3 \\ y_3' &= & & 2y_3 \end{cases}$$

with initial conditions: $y_1(0) = y_2(0) = y_3(0) = 1$. Let

$$\mathbf{A} = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ 0 & 0 & 2 \end{pmatrix}, \text{ eigenvalues are } \lambda = 2, 4.$$

A basis for E_2 is $\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$, and a basis for E_4 is $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}$

$\Rightarrow \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^{2t}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} e^{2t}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} e^{4t}$ are solutions to the SDE.

Example

Wronskian is

$$W\left(e^{2t}\begin{pmatrix}1\\0\\1\end{pmatrix}, e^{2t}\begin{pmatrix}0\\1\\1\end{pmatrix}, e^{4t}\begin{pmatrix}1\\1\\0\end{pmatrix}\right) = (e^{2t})(e^{2t})(e^{4t})\begin{vmatrix}1 & 0 & 1\\0 & 1 & 1\\1 & 1 & 0\end{vmatrix} = -2e^{8t} \neq 0.$$

Hence, the general solution is $c_1\begin{pmatrix}1\\0\\1\end{pmatrix}e^{2t} + c_2\begin{pmatrix}0\\1\\1\end{pmatrix}e^{2t} + c_3\begin{pmatrix}1\\1\\0\end{pmatrix}e^{4t},\ c_1, c_2, c_3 \in \mathbb{R}.$

Solve for initial conditions

$$\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}1 & 0 & 1\\0 & 1 & 1\\1 & 1 & 0\end{pmatrix}\begin{pmatrix}c_1\\c_2\\c_3\end{pmatrix}, \left(\begin{array}{ccc|c}1 & 0 & 1 & 1\\0 & 1 & 1 & 1\\1 & 1 & 0 & 1\end{array}\right) \xrightarrow{RREF} \left(\begin{array}{ccc|c}1 & 0 & 0 & 1/2\\0 & 1 & 0 & 1/2\\0 & 0 & 1 & 1/2\end{array}\right)$$

So, the solution to the initial value problem is

$$\mathbf{x}(t) = \frac{e^{2t}}{2}\begin{pmatrix}1\\0\\1\end{pmatrix} + \frac{e^{2t}}{2}\begin{pmatrix}0\\1\\1\end{pmatrix} + \frac{e^{4t}}{2}\begin{pmatrix}1\\1\\0\end{pmatrix}.$$

Non-diagonalizable?

Recall that there are two obstructions to diagonalization,

(i) the characteristic polynomial does not factorize into real linear factors,

Solution: allow complex eigenvalues

(ii) the geometric multiplicity of an eigenvalue is strictly less than the algebraic multiplicity.

Solution: use generalized eigenvectors

7.2 Complex Eigenvalues and Eigenvectors

Complex vectors

Definition

An *n*-complex vector is a collection of n ordered complex numbers,

$$\mathbf{v} = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{pmatrix}, z_j \in \mathbb{C} \text{ for all } j = 1, ..., n.$$

The collection of all n-complex vectors is denoted as $\mathbb{C}^n$. Given any complex vector $\mathbf{v} \in \mathbb{C}^n$, we can split it into its real and imaginary parts,

$$\mathbf{v} = \begin{pmatrix} x_1 + iy_1 \\ x_2 + iy_2 \\ \vdots \\ x_n + iy_n \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + i \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = Re(\mathbf{v}) + iIm(\mathbf{v}),$$

where $Re(\mathbf{v}), Im(\mathbf{v}) \in \mathbb{R}^n$.

Complex eigenvalues and eigenvectors

Definition

Let $\mathbf{A}$ be an order n real square matrix. $\lambda \in \mathbb{C}$ is an eigenvalue of $\mathbf{A}$ if there is a nonzero vector $\mathbf{v} \in \mathbb{C}^n$, $\mathbf{v} \neq \mathbf{0}$ such that

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

In this case, $\mathbf{v}$ is called an eigenvector associated to λ.

Examples

1. Let $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} i \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ i \end{pmatrix} = i \begin{pmatrix} i \\ 1 \end{pmatrix}.$$

   So, $\lambda = i$ is an eigenvalue of $\mathbf{A}$ with associated eigenvector $\begin{pmatrix} i \\ 1 \end{pmatrix}$.

2. Let $\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ -1 & 2 & 1 \end{pmatrix}$.

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ -1 & 2 & 1 \end{pmatrix} \begin{pmatrix} -i \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1-i \\ 0 \\ 1+i \end{pmatrix} = (1+i) \begin{pmatrix} -i \\ 0 \\ 1 \end{pmatrix}.$$

   So, $\lambda = 1 + i$ is an eigenvalue of $\mathbf{A}$ with associated eigenvector $\begin{pmatrix} -i \\ 0 \\ 1 \end{pmatrix}$.

Complex Conjugate

Definition

For a complex number $z = x + iy$, let $\overline{z} = x - iy$ denote its complex conjugate. Similarly, for a complex vector

$$\mathbf{v} = \begin{pmatrix} x_1 + iy_1 \\ x_2 + iy_2 \\ \vdots \\ x_n + iy_n \end{pmatrix}, \text{ let } \overline{\mathbf{v}} = \begin{pmatrix} x_1 - iy_1 \\ x_2 - iy_2 \\ \vdots \\ x_n - iy_n \end{pmatrix}$$

denote its complex conjugate.

Examples

1. $\mathbf{v} = \begin{pmatrix} i \\ 1 \end{pmatrix}$, $\overline{\mathbf{v}} = \begin{pmatrix} -i \\ 1 \end{pmatrix}$.

2. $\mathbf{v} = \begin{pmatrix} 1-i \\ 0 \\ 1+i \end{pmatrix}$, $\overline{\mathbf{v}} = \begin{pmatrix} 1+i \\ 0 \\ 1-i \end{pmatrix}$.

Complex eigenvalues and eigenvectors

Theorem

Let $\mathbf{A}$ *be an order* n *square matrix with real entries.*

(i) *Then complex eigenvalues of* $\mathbf{A}$ *comes in conjugate pairs, that is, if* $\lambda \in \mathbb{C}$ *is an eigenvalue of* $\mathbf{A}$*, then* $\overline{\lambda}$ *is also an eigenvalue of* $\mathbf{A}$*.*

(ii) *If* $\mathbf{v} \in \mathbb{C}^n$ *is an eigenvector associated to eigenvalue* λ*, then* $\overline{\mathbf{v}}$ *is an eigenvector associated to eigenvalue* $\overline{\lambda}$*.*

Proof.

Suppose λ is a (complex) eigenvalue of a real-valued matrix $\mathbf{A}$ associated to eigenvector $\mathbf{v}$. Then, since $\mathbf{A}$ is real-value, $\overline{\mathbf{A}} = \mathbf{A}$. Hence,

$$\mathbf{A}\overline{\mathbf{v}} = \overline{\mathbf{A}\mathbf{v}} = \overline{\lambda\mathbf{v}} = \overline{\lambda}\overline{\mathbf{v}},$$

which shows that $\overline{\lambda}$ is an eigenvalue of $\mathbf{A}$ associated to eigenvector $\overline{\mathbf{v}}$. $\square$

Example

We have seen that $\lambda = i$ is an eigenvalue of $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ associated to the eigenvector $\mathbf{v} = \begin{pmatrix} i \\ 1 \end{pmatrix}$. Now

$$\mathbf{A}\overline{\mathbf{v}} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} -i \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -i \end{pmatrix} = -i \begin{pmatrix} -i \\ 1 \end{pmatrix}.$$

So, $\overline{\mathbf{v}} = \begin{pmatrix} -i \\ 1 \end{pmatrix}$ is an eigenvector of $\mathbf{A}$ associated to eigenvalue $\overline{i} = -i$.

We have seen that $\lambda = 1 + i$ is an eigenvalue of $\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ -1 & 2 & 1 \end{pmatrix}$ associated to eigenvector $\mathbf{v} = \begin{pmatrix} -i \\ 0 \\ 1 \end{pmatrix}$. Now

$$\mathbf{A}\overline{\mathbf{v}} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ -1 & 2 & 1 \end{pmatrix} \begin{pmatrix} i \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+i \\ 0 \\ 1-i \end{pmatrix} = (1-i) \begin{pmatrix} i \\ 0 \\ 1 \end{pmatrix}$$

So, $\overline{1+i} = 1 - i$ is also an eigenvalue of $\mathbf{A}$ with eigenvector $\overline{\mathbf{v}} = \begin{pmatrix} i \\ 0 \\ 1 \end{pmatrix}$.

Example

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ -1 & 2 & 1 \end{pmatrix}. \quad char(\mathbf{A}) = \begin{vmatrix} x-1 & 0 & -1 \\ 0 & x-2 & 0 \\ 1 & -2 & x-1 \end{vmatrix} = x^3 - 4x^2 + 6x - 4 = (x-2)(x-(1+i))(x-(1-i)).$$

Find the eigenvectors:

$$\lambda = 2: \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & -2 & 1 \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

$$\lambda = 1+i: \begin{pmatrix} i & 0 & -1 \\ 0 & -1+i & 0 \\ 1 & -2 & i \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & 0 & i \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \mathbf{v}_{1+i} = \begin{pmatrix} -i \\ 0 \\ 1 \end{pmatrix}.$$

$$\lambda = 1-i: \begin{pmatrix} -i & 0 & -1 \\ 0 & -1-i & 0 \\ 1 & -2 & -i \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & 0 & -i \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \mathbf{v}_{1-i} = \begin{pmatrix} i \\ 0 \\ 1 \end{pmatrix}.$$

Complex Solutions to Real SDE

Theorem

Suppose $\lambda \in \mathbb{C}$ is a complex (nonreal) eigenvalue of $\mathbf{A}$ (of order n) with eigenvector $\mathbf{v} \in \mathbb{C}^n$. Then $\mathbf{v}e^{\lambda t}$ is a complex solution to the differential system $\mathbf{y}' = \mathbf{A}\mathbf{y}$.

Proof.

Extend verbatim from the real eigenvalue and eigenvector case. $\square$

Remark: However, complex solutions do not solve real problems.

Question

1. Which of the following identities correctly represents e^{x+iy} for real numbers x and y?
   (i) $e^{x+iy} = \cos(x+iy) + \sin(x+iy)$
   (ii) $e^{x+iy} = \cos(x) + i\sin(y)$
   (iii) $e^{x+iy} = e^x(\cos y + i\sin y)$
   (iv) $e^{x+iy} = x(\cos y + i\sin y)$

2. Which of the following correctly represents the product of two complex numbers $(x_1 + iy_1)$ and $(x_2 + iy_2)$ for real numbers x_1, y_1, x_2, y_2?
   (i) $(x_1 + iy_1)(x_2 + iy_2) = x_1x_2 + iy_1y_2$
   (ii) $(x_1 + iy_1)(x_2 + iy_2) = x_1x_2 - y_1y_2 + i(x_1y_2 + x_2y_1)$
   (iii) $(x_1 + iy_1)(x_2 + iy_2) = x_1x_2 + y_1y_2 + i(x_1y_2 + x_2y_1)$
   (iv) $(x_1 + iy_1)(x_2 + iy_2) = x_1x_2 + y_1y_2 + i(x_1y_2 - x_2y_1)$

Real solutions from complex solution

Suppose $\lambda \in \mathbb{C}$ is a complex eigenvalue of $\mathbf{A}$ associated to complex eigenvector $\mathbf{v} \in \mathbb{C}^n$. Decompose λ and $\mathbf{v}$ into their real and imaginary parts,

$$\lambda = \lambda_r + i\lambda_i, \mathbf{v} = \mathbf{v}_r + i\mathbf{v}_i, \qquad \lambda_r, \lambda_i \in \mathbb{R}, \quad \mathbf{v}_r, \mathbf{v}_i \in \mathbb{R}^n.$$

Then

$$\begin{aligned} e^{\lambda t}\mathbf{v} &= e^{(\lambda_r + i\lambda_i)t}(\mathbf{v}_r + i\mathbf{v}_i) \\ &= e^{\lambda_r t}(\cos(\lambda_i t) + i\sin(\lambda_i t))(\mathbf{v}_r + i\mathbf{v}_i) \\ &= e^{\lambda_r t}(\cos(\lambda_i t)\mathbf{v}_r - \sin(\lambda_i t)\mathbf{v}_i) + ie^{\lambda_r t}(\sin(\lambda_i t)\mathbf{v}_r + \cos(\lambda_i t)\mathbf{v}_i) \\ &= \mathbf{x}_r(t) + i\mathbf{x}_i(t), \end{aligned}$$

where

$$Re(e^{\lambda t}\mathbf{v}) = \mathbf{x}_r(t) = e^{\lambda_r t}(\cos(\lambda_i t)\mathbf{v}_r - \sin(\lambda_i t)\mathbf{v}_i)$$

and

$$Im(e^{\lambda t}\mathbf{v}) = \mathbf{x}_i(t) = e^{\lambda_r t}(\sin(\lambda_i t)\mathbf{v}_r + \cos(\lambda_i t)\mathbf{v}_i)$$

are real function-valued vectors.

Real solutions from complex solution

Theorem

Both $\mathbf{x}_r(t)$ and $\mathbf{x}_i(t)$ are real solutions to the differential equations $\mathbf{y}' = \mathbf{A}\mathbf{y}$. Moreover, $\{\mathbf{x}_r(t)$ and $\mathbf{x}_i(t)\}$ are linearly independent.

Proof.

We will only show that $\mathbf{x}_r(t)$ and $\mathbf{x}_i(t)$ are solutions; the proof of linearly independence is omitted. Let $\lambda \in \mathbb{C}$ be a nonreal eigenvalue of $\mathbf{A}$ and $\mathbf{v}$ an associated eigenvector. Then $\mathbf{x}(t) = \mathbf{v}e^{\lambda t}$ is a complex solution to the SDE. Write

$$\mathbf{x}(t) = \mathbf{x}_r(t) + i\mathbf{x}_i(t).$$

Then

$$\mathbf{A}\mathbf{x}_r(t) + i\mathbf{A}\mathbf{x}_i(t) = \mathbf{A}(\mathbf{x}_r(t) + i\mathbf{x}_i(t)) = \mathbf{A}\mathbf{x}(t) = \mathbf{x}'(t) = \mathbf{x}_r'(t) + i\mathbf{x}_i'(t).$$

By comparing the real and imaginary part, we have for all $t \in \mathbb{R}$,

$$\mathbf{A}\mathbf{x}_r(t) = \mathbf{x}_r'(t) \quad \text{and} \quad \mathbf{A}\mathbf{x}_i(t) = \mathbf{x}_i'(t),$$

which show that $\mathbf{x}_r(t)$ and $\mathbf{x}_i(t)$ are solutions. $\square$

Real solutions from complex solution

Theorem

Suppose $\mathbf{A}$ *is an order* 2 *square matrix and* $\lambda \in \mathbb{C}$ *is a nonreal complex eigenvalue* $\lambda \notin \mathbb{R}$. *Let* $\mathbf{v} \in \mathbb{C}^2$ *be an associated eigenvector. Write* $e^{\lambda t}\mathbf{v} = \mathbf{x}_r(t) + i\mathbf{x}_i(t)$. *Then* $\{\mathbf{x}_r(t), \mathbf{x}_i(t)\}$ *is a fundamental set of solutions for* $\mathbf{y}' = \mathbf{A}\mathbf{y}$, *and*

$$\mathbf{x} = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t)$$

is a general solution.

Here

$$\mathbf{x}_r(t) = e^{\lambda_r t}(\cos(\lambda_i t)\mathbf{v}_r - \sin(\lambda_i t)\mathbf{v}_i)$$

and

$$\mathbf{x}_i(t) = e^{\lambda_r t}(\sin(\lambda_i t)\mathbf{v}_r + \cos(\lambda_i t)\mathbf{v}_i),$$

where

$$\lambda = \lambda_r + i\lambda_i, \mathbf{v} = \mathbf{v}_r + i\mathbf{v}_i, \lambda_r, \lambda_i \in \mathbb{R}, \mathbf{v}_r, \mathbf{v}_i \in \mathbb{R}^n.$$

Example

Solve the following differential system

$$\begin{array}{rcrcr} y_1' & = & & - & y_2 \\ y_2' & = & y_1 & & \end{array}$$

with initial conditions $y_1(0) = 1 = y_2(0)$.

The coefficient matrix is $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. $char(\mathbf{A}) = \begin{vmatrix} x & 1 \\ -1 & x \end{vmatrix} = x^2 + 1$. The eigenvalues are $\lambda = \pm i$. We will work with eigenvalue $\lambda = i$.

$$i\mathbf{I} - \mathbf{A} = \begin{pmatrix} i & 1 \\ -1 & i \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & -i \\ 0 & 0 \end{pmatrix} \quad \Rightarrow \quad \mathbf{v} = \begin{pmatrix} i \\ 1 \end{pmatrix}.$$

Let $\lambda_r = 0$, $\lambda_i = 1$, $\mathbf{v}_r = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, and $\mathbf{v}_i = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. So,

$$\begin{aligned} \mathbf{x}_r(t) &= e^{0t}\left(\cos(t)\begin{pmatrix} 0 \\ 1 \end{pmatrix} - \sin(t)\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} -\sin(t) \\ \cos(t) \end{pmatrix} \\ \mathbf{x}_i(t) &= e^{0t}\left(\sin(t)\begin{pmatrix} 0 \\ 1 \end{pmatrix} + \cos(t)\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} \cos(t) \\ \sin(t) \end{pmatrix} \end{aligned}$$

Example

Check that $\left\{ \begin{pmatrix} -\sin(t) \\ \cos(t) \end{pmatrix}, \begin{pmatrix} \cos(t) \\ \sin(t) \end{pmatrix} \right\}$ is a fundamental set of solutions.

$$\text{Wronskian: } \begin{vmatrix} -\sin(t) & \cos(t) \\ \cos(t) & \sin(t) \end{vmatrix} = -\sin^2(t) - \cos^2(t) = -1 \neq 0.$$

General solution: $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = c_1 \begin{pmatrix} -\sin(t) \\ \cos(t) \end{pmatrix} + c_2 \begin{pmatrix} \cos(t) \\ \sin(t) \end{pmatrix}$.

Now solve for the initial condition

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -c_1 \sin(0) + c_2 \cos(0) \\ c_1 \cos(0) + c_s \sin(0) \end{pmatrix} = \begin{pmatrix} c_2 \\ c_1 \end{pmatrix} \quad \Rightarrow \quad c_1 = c_2 = 1$$

So solution to the initial value problem is

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} -\sin(t) \\ \cos(t) \end{pmatrix} + \begin{pmatrix} \cos(t) \\ \sin(t) \end{pmatrix}, \textit{or} \begin{cases} y_1(t) &= \cos(t) - \sin(t) \\ y_2(t) &= \cos(t) + \sin(t). \end{cases}$$

Example

Solve the following differential system

$$\begin{array}{rcrcrcr} y_1' & = & y_1 & & & + & y_3 \\ y_2' & = & & & 2y_2 & & \\ y_3' & = & -y_1 & + & 2y_2 & + & y_3 \end{array}$$

The eigenvalues are $\lambda = 2, 1+i, 1-i$ with associated eigenvectors $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\mathbf{v}_{1+i} = \begin{pmatrix} -i \\ 0 \\ 1 \end{pmatrix}$, $\mathbf{v}_{1-i} = \begin{pmatrix} i \\ 0 \\ 1 \end{pmatrix}$, respectively.

- Using $\lambda = 2$ and its associated eigenvector, we have that $\mathbf{x}_1(t) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{2t}$ is a solution.
- Using $\lambda = 1+i$ and its associated eigenvector, we have $\lambda_r = 1$, $\lambda_i = 1$, $\mathbf{v}_r = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$, and $\mathbf{v}_i = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$, we have that $\mathbf{x}_r(t) = e^t \begin{pmatrix} \sin(t) \\ 0 \\ \cos(t) \end{pmatrix}$, and $\mathbf{x}_i(t) = e^t \begin{pmatrix} -\cos(t) \\ 0 \\ \sin(t) \end{pmatrix}$ are solutions

Example

Check that $\{\mathbf{x}_1(t), \mathbf{x}_r(t), \mathbf{x}_i(t)\}$ is a fundamental set of solutions.

$$W\left(\begin{pmatrix}1\\1\\1\end{pmatrix}e^{2t}, e^t\begin{pmatrix}\sin(t)\\0\\\cos(t)\end{pmatrix}, e^t\begin{pmatrix}-\cos(t)\\0\\\sin(t)\end{pmatrix}\right) = e^{4t}\begin{vmatrix}1 & \sin(t) & -\cos(t)\\1 & 0 & 0\\1 & \cos(t) & \sin(t)\end{vmatrix} = -e^{4t}(\sin^2(t)+\cos^2(t)) = -e^{4t} \neq 0.$$

Hence, the general solution to the SDE is

$$c_1 e^{2t}\begin{pmatrix}1\\1\\1\end{pmatrix} + c_2 e^t\begin{pmatrix}\sin(t)\\0\\\cos(t)\end{pmatrix} + c_3 e^t\begin{pmatrix}-\cos(t)\\0\\\sin(t)\end{pmatrix}.$$

7.3 Repeated Eigenvalue and Generalized Eigenvector

Repeated eigenvalue and generalized eigenvector

Definition

Let $\mathbf{A}$ be an order n square matrix. An eigenvalue λ of $\mathbf{A}$ is a repeated eigenvalue if the algebraic multiplicity of λ is more than 1,

$$r_\lambda > 1.$$

Suppose now λ is a repeated eigenvalue such that the geometric multiplicity is strictly smaller than the algebraic multiplicity,

$$\dim(E_\lambda) < r_\lambda.$$

In this case, we do not have enough linearly independent eigenvectors associated to eigenvalue λ to diagonalize $\mathbf{A}$, and hence do not have enough linearly independent solutions to form a fundamental set of solutions. To address this, we introduce the concept of a generalized eigenvector.

Let $\mathbf{v}_1$ be an eigenvector associated to λ. A vector $\mathbf{v}_2$ is a generalized eigenvector associated to λ if it is a solution to

$$(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{v}_1.$$

Example

Let $\mathbf{A} = \begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}$. $char(\mathbf{A}) = \begin{vmatrix} x+1 & -1 \\ 4 & x-3 \end{vmatrix} = (x-1)^2$. The only eigenvalue if $\lambda = 1$, with algebraic multiplicity $r_1 = 2$.

$$\mathbf{I} - \mathbf{A} = \begin{pmatrix} 2 & -1 \\ 4 & -2 \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & -1/2 \\ 0 & 0 \end{pmatrix} \quad \Rightarrow \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

This shows that $\lambda = 1$ is a repeated eigenvalue. Now find a generalized eigenvector, solve for $(\mathbf{A} - \mathbf{I})x = \mathbf{v}_1$

$$\left(\begin{array}{cc|c} -2 & 1 & 1 \\ -4 & 2 & 2 \end{array}\right) \xrightarrow{RREF} \left(\begin{array}{cc|c} 1 & -1/2 & -1/2 \\ 0 & 0 & 0 \end{array}\right)$$

So, any vector of the form

$$\begin{pmatrix} -1/2 \\ 0 \end{pmatrix} + s \begin{pmatrix} 1/2 \\ 1 \end{pmatrix}, \quad s \in \mathbb{R}$$

is a generalized eigenvector associated to $\lambda = 1$. For example, let $s = 1$, then $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is a generalized eigenvector.

Solutions from repeated eigenvalue and generalized eigenvector

Let λ be a repeated eigenvalue of an order n matrix $\mathbf{A}$, with eigenvector $\mathbf{v}_1$ and a generalized eigenvector $\mathbf{v}_2$ associated to λ. Let

$$\begin{aligned} \mathbf{x}_1(t) &= e^{\lambda t}\mathbf{v}_1 \\ \mathbf{x}_2(t) &= e^{\lambda t}(t\mathbf{v}_1 + \mathbf{v}_2). \end{aligned}$$

Then $\{\mathbf{x}_1(t), \mathbf{x}_2(t)\}$ is a linearly independent set of real solutions to the differential equations $\mathbf{y}' = \mathbf{A}\mathbf{y}$.

Theorem

Let $\mathbf{A}$ be an order 2 square matrix with a repeated eigenvalue λ. Suppose $\mathbf{v}_1$ is an eigenvector, and $\mathbf{v}_2$ a generalized eigenvector associated to λ. Then the set

$$\{\mathbf{x}_1(t), \mathbf{x}_2(t)\}$$

forms a fundamental set of solutions for $\mathbf{y}' = \mathbf{A}\mathbf{y}$.

Example

Solve the differential equations

$$\begin{array}{rcrcr} y_1' & = & -y_1 & + & y_2 \\ y_2' & = & -4y_1 & + & 3y_2 \end{array}$$

with initial conditions $y_1(0) = 2$, $y_2(0) = 1$.

$\lambda = 1$ is a repeated eigenvalue with eigenvector $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and generalized eigenvector $\mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. The two solutions are

$$\begin{aligned} \mathbf{x}_1(t) &= e^{\lambda t}\mathbf{v}_1 = e^t \begin{pmatrix} 1 \\ 2 \end{pmatrix} \\ \mathbf{x}_2(t) &= e^{\lambda t}(t\mathbf{v}_1 + \mathbf{v}_2) = e^t \begin{pmatrix} t \\ 2t+1 \end{pmatrix} \end{aligned}$$

Check that they are independent

$$W\left(e^t \begin{pmatrix} 1 \\ 2 \end{pmatrix}, e^t \begin{pmatrix} t \\ 2t+1 \end{pmatrix}\right) = e^{2t} \begin{vmatrix} 1 & t \\ 2 & 2t+1 \end{vmatrix} = e^{2t} \neq 0.$$

Example

So,

Fundamental set of solution: $\left\{ e^t \begin{pmatrix} 1 \\ 2 \end{pmatrix}, e^t \begin{pmatrix} t \\ 2t+1 \end{pmatrix} \right\}$

General solution: $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = c_1 e^t \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 e^t \begin{pmatrix} t \\ 2t+1 \end{pmatrix}, \quad c_1, c_2 \in \mathbb{R}.$

Solve the initial conditions

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} c_1 \\ 2c_1 + c_2 \end{pmatrix} \quad \Rightarrow \quad c_1 = 2, c_2 = -3.$$

Solution to the initial value problem

$$\begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix} = 2e^t \begin{pmatrix} 1 \\ 2 \end{pmatrix} - 3e^t \begin{pmatrix} t \\ 2t+1 \end{pmatrix} = e^t \begin{pmatrix} 2-3t \\ 1-6t \end{pmatrix} \quad \text{or} \quad \begin{cases} y_1(t) &= e^t(2-3t) \\ y_2(t) &= e^t(1-6t) \end{cases}$$

Example

Solve the following differential equations

$$\begin{array}{rcrcrcr} y_1' & = & 3y_1 & - & y_2 & & \\ y_2' & = & 4y_1 & - & y_2 & & \\ y_3' & = & 2y_1 & - & y_2 & + & 2y_3 \end{array}$$

The eigenvalues of $\mathbf{A} = \begin{pmatrix} 3 & -1 & 0 \\ 4 & -1 & 0 \\ 2 & -1 & 2 \end{pmatrix}$ are $\lambda = 1, 2$ with multiplicities $r_2 = 1, r_1 = 2$.

$$\mathbf{I} - \mathbf{A} = \begin{pmatrix} -2 & 1 & 0 \\ -4 & 2 & 0 \\ -2 & 1 & -1 \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & -1/2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow E_1 = span\left\{ \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right\} \Rightarrow \dim(E_1) = 1 < 2 = r_1$$

So, $\lambda = 1$ is a repeated eigenvalue.

Example

Find generalized eigenvector, solve for $(\mathbf{A} - \mathbf{I})\mathbf{x} = \mathbf{v}_1$:

$$\left(\begin{array}{ccc|c} 2 & -1 & 0 & 1 \\ 4 & -2 & 0 & 2 \\ 2 & -1 & 1 & 0 \end{array}\right) \xrightarrow{RREF} \left(\begin{array}{ccc|c} 1 & -1/2 & 0 & 1/2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

So, any vector of the form

$$\begin{pmatrix} 1/2 \\ 0 \\ -1 \end{pmatrix} + s \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix}, \quad s \in \mathbb{R}$$

is a generalized eigenvector. In particular, let $s = -1$, we have $\mathbf{v}_2 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$.

Example

Now find an eigenvector associated to $\lambda = 2$.

$$2\mathbf{I} - \mathbf{A} = \begin{pmatrix} -1 & 1 & 0 \\ -4 & 3 & 0 \\ -2 & 1 & 0 \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \mathbf{u} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

So,

$$e^{2t}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad e^{t}\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad e^{t}\left(t\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}\right) = e^{t}\begin{pmatrix} t \\ 2t-1 \\ 1 \end{pmatrix}$$

are solutions to the SDE. Check that they are independent.

$$W\left(e^{2t}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, e^{t}\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, e^{t}\begin{pmatrix} t \\ 2t-1 \\ 1 \end{pmatrix}\right) = e^{4t}\begin{vmatrix} 0 & 1 & t \\ 0 & 2 & 2t-1 \\ 1 & 0 & 1 \end{vmatrix} = -e^{4t} \neq 0.$$

Example

Therefore, the general solution is

$$c_1 e^{2t} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + c_2 e^{t} \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + c_3 e^{t} \begin{pmatrix} t \\ 2t-1 \\ 1 \end{pmatrix}, \quad c_1, c_2 c_3 \in \mathbb{R}.$$

Example

Consider the differential system

$$\begin{cases} y_1' &= 4y_1 - 4y_2 - 2y_3 \\ y_2' &= y_2 \\ y_3' &= 2y_1 - 3y_2 \end{cases}.$$

Characteristic polynomial: $\begin{vmatrix} x-4 & 4 & 2 \\ 0 & x-1 & 0 \\ -2 & 3 & x \end{vmatrix} = x^3 - 5x^2 + 8x - 4 = (x-1)(x-2)^2$. The eigenvalues are $\lambda = 1, 2$, with algebraic multiplicities $r_1 = 1, r_2 = 2$.

Find the eigenvectors associated to eigenvalue 2.

$$2\mathbf{I} - \mathbf{A} = \begin{pmatrix} -2 & 4 & 2 \\ 0 & 1 & 0 \\ -2 & 3 & 2 \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

An eigenvector is $\mathbf{v} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$. Since $\dim(E_2) = 1 < 2 = r_2$, $\lambda = 2$ is a repeated eigenvalue.

Example

Find a generalized eigenvector.

$$(\mathbf{A} - 2\mathbf{I} \mid \mathbf{v}) = \left(\begin{array}{ccc|c} 2 & -4 & -2 & 1 \\ 0 & -1 & 0 & 0 \\ 2 & -3 & -2 & 1 \end{array}\right) \xrightarrow{RREF} \left(\begin{array}{ccc|c} 1 & 0 & -1 & 1/2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

So, a generalized eigenvector is $\begin{pmatrix} 1/2 \\ 0 \\ 0 \end{pmatrix}$.

Finally, find an eigenvector associated to eigenvalue 1.

$$\mathbf{I} - \mathbf{A} = \begin{pmatrix} -3 & 4 & 2 \\ 0 & 0 & 0 \\ -2 & 3 & 1 \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

An eigenvector is $\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$.

Example

Hence,

$$e^t \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, e^{2t} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, e^{2t} \left(\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} t + \begin{pmatrix} 1/2 \\ 0 \\ 0 \end{pmatrix} \right)$$

are solutions to the differential system. Now check that they are linearly independent. The Wronskian is

$$W(0) = W\left(\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1/2 \\ 0 \\ 0 \end{pmatrix} \right) = \begin{vmatrix} 2 & 1 & 1/2 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{vmatrix} = 1/2 \neq 0.$$

Hence,

$$\left\{ e^t \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, e^{2t} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, e^{2t} \left(\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} t + \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \right) \right\}$$

is a fundamental set of solutions.

Example

Let $y_1(0) = y_2(0) = y_3(0) = 1$ be the initial condition. The solution to the initial value problem is

$$\begin{pmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{pmatrix} = e^t \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} - e^{2t} \begin{pmatrix} 1+2t \\ 0 \\ 2t \end{pmatrix}.$$

Example

Solve the initial value problem

$$\begin{cases} y_1' &= y_1 & & + 2y_3 & \\ y_2' &= -y_1 - y_2 & & + 2y_3 & - y_4 \\ y_3' &= -2y_1 & & + y_3 & \\ y_4' &= 2y_1 & & + 2y_3 & - y_4 \end{cases},$$

with initial conditions $y_1(0) = y_2(0) = y_3(0) = y_4(0) = 1$. The characteristic polynomial is $\begin{vmatrix} x-1 & 0 & -2 & 0 \\ 1 & x+1 & -2 & 1 \\ 2 & 0 & x-1 & 0 \\ -2 & 0 & -2 & x+1 \end{vmatrix} = x^4 + 2x^2 + 8x + 5 = (x+1)^2(x-(1+2i))(x-(1-2i))$. The eigenvalues are $\lambda = -1, 1 \pm 2i$, with algebraic multiplicity $r_{-1} = 2, r_{1\pm 2i} = 1$.

Example

Find the eigenvectors associated to eigenvalue 1.

$$(-\mathbf{I}-\mathbf{A}) = \begin{pmatrix} -2 & 0 & -2 & 0 \\ 1 & 0 & -2 & 1 \\ 2 & 0 & -2 & 0 \\ -2 & 0 & -2 & 0 \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

So, an eigenvector is $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$. Since $\dim(E_1) = 1 < 2 = r_{-1}$, we need to find a generalized eigenvector associated to $\lambda = 1$.

$$(\mathbf{A}+\mathbf{I} \mid \mathbf{v}) = \left(\begin{array}{cccc|c} 2 & 0 & 2 & 0 & 0 \\ -1 & 0 & 2 & -1 & 1 \\ -2 & 0 & 2 & 0 & 0 \\ 2 & 0 & 2 & 0 & 0 \end{array}\right) \xrightarrow{RREF} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

A generalized eigenvector is $\begin{pmatrix} 0 \\ 0 \\ 0 \\ -1 \end{pmatrix}$. So, $e^{-t}\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$ and $e^{-t}\begin{pmatrix} 0 \\ t \\ 0 \\ -1 \end{pmatrix}$ are solutions to the differential systems.

Example

Now consider $\lambda = 1 + 2i$. Find an associated eigenvector.

$$(1+2i)\mathbf{I} - \mathbf{A} = \begin{pmatrix} 2i & 0 & -2 & 0 \\ 1 & 2+2i & -2 & 1 \\ 2 & 0 & 2i & 0 \\ -2 & 0 & -2 & 2+2i \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1i \\ 0 & 0 & 1 & -1i \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

An eigenvector is $\begin{pmatrix} 1 \\ i \\ i \\ 1 \end{pmatrix}$. Then $\lambda_r = 1, \lambda_i = 2, \mathbf{v}_r = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \mathbf{v}_i = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}$. So let $\mathbf{x}_r = e^t \begin{pmatrix} \cos(2t) \\ -\sin(2t) \\ -\sin(2t) \\ \cos(2t) \end{pmatrix}$ and $\mathbf{x}_i = e^t \begin{pmatrix} \sin(2t) \\ \cos(2t) \\ \cos(2t) \\ \sin(2t) \end{pmatrix}$ are solutions to the differential systems.

Example

Now check that

$$\left\{ e^{-t}\begin{pmatrix}0\\1\\0\\0\end{pmatrix}, e^{-t}\begin{pmatrix}0\\t\\0\\-1\end{pmatrix}, e^{t}\begin{pmatrix}\cos(2t)\\-\sin(2t)\\-\sin(2t)\\\cos(2t)\end{pmatrix}, e^{t}\begin{pmatrix}\sin(2t)\\\cos(2t)\\\cos(2t)\\\sin(2t)\end{pmatrix}\right\}$$

is linearly independent. Indeed,

$$W(0) = W\left(\begin{pmatrix}0\\1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\0\\0\\-1\end{pmatrix}, \begin{pmatrix}1\\0\\0\\1\end{pmatrix}, \begin{pmatrix}0\\1\\1\\0\end{pmatrix}\right) = \begin{vmatrix}0 & 0 & 1 & 0\\1 & 0 & 0 & 1\\0 & 0 & 0 & 1\\0 & -1 & 1 & 0\end{vmatrix} = 1 \neq 0.$$

The general solution is

$$c_1 e^{-t}\begin{pmatrix}0\\1\\0\\0\end{pmatrix} + c_2 e^{-t}\begin{pmatrix}0\\t\\0\\-1\end{pmatrix} + c_3 e^{t}\begin{pmatrix}\cos(2t)\\-\sin(2t)\\-\sin(2t)\\\cos(2t)\end{pmatrix} + c_4 e^{t}\begin{pmatrix}\sin(2t)\\\cos(2t)\\\cos(2t)\\\sin(2t)\end{pmatrix}, \quad c_1, c_2, c_3, c_4 \in \mathbb{R}.$$

Example

Solve the initial conditions.

$$\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} c_3 \\ c_1 + c_4 \\ c_4 \\ -c_2 + c_4 \end{pmatrix}$$

tells us that $c_1 = c_2 = 0$, $c_3 = c_4 = 1$. Hence, the solution to the initial value problem is

$$\begin{pmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \\ y_4(t) \end{pmatrix} = e^t \begin{pmatrix} \cos(2t) + \sin(2t) \\ \cos(2t) - \sin(2t) \\ \cos(2t) - \sin(2t) \\ \cos(2t) + \sin(2t) \end{pmatrix}.$$